

Some basic concepts of chemistry : DPP

Chemistry

Vishal Kumar Sir

- Q1** Which of the following statement is/are incorrect?
(A) All heterogeneous mixtures are called as solution.
(B) Element can not be broken down into simple substances.
(C) All homogeneous mixtures are called as solution.
(D) A compound can be decomposed into its components.
- Q2** After a chemical reaction, the total mass of reactants and products
(A) Is always increased
(B) Is always decreased
(C) Is not changed
(D) Is always less or more
- Q3** The law of conservation of mass is valid for all the following, except
(A) All chemical reactions
(B) Nuclear reactions
(C) Endothermic reactions
(D) Exothermic reactions
- Q4** Which of the following is the best example of law of conservation of mass
(A) 12 g of carbon combines with 32 g of oxygen to form 44 g of CO_2
(B) When 12 g of carbon is heated in a vacuum there is no change in mass
(C) A sample of air increases in volume when heated at constant pressure but its mass remains unaltered
(D) The weight of a piece of platinum is the same before and after heating in air
- Q5** 10g CaCO_3 on heating gives 5.6 g CaO and g CO_2 .
(A) 4.4 (B) 5.6
(C) 6.5 (D) 4.2
- Q6** Among the following pairs of compounds, the one that illustrates the law of multiple proportions is
(A) NH_3 and NCl_3
(B) H_2S and SO_2
(C) CuO and Cu_2O
(D) CS_2 and FeSO_4
- Q7** An element forms two oxides containing respectively 53.33 and 36.36 percent of oxygen. These figures illustrate the law of:
(A) Conservation of mass
(B) Reciprocal proportions
(C) Constant proportions
(D) Multiple proportions
- Q8** Atomic mass of boron is 10.81. It has two isotopes with 80% and 20% abundance respectively. The atomic mass of the isotope having 80% abundance is 11.01. The atomic mass of the other isotope is
(A) 10.80 (B) 11.01
(C) 10.01 (D) 21.82
- Q9** Carbon occurs in nature as a mixture of C^{12} and C^{13} . Average atomic mass of carbon is 12.011. What is the % abundance of C^{12} in nature?
(A) 99.8% (B) 98.9%
(C) 97.6% (D) 98.5%



[Android App](#)

| [iOS App](#)

| [PW Website](#)

Q10 One mole electron means:

- (A) N_A electrons
- (B) 6.023×10^{23} electrons
- (C) 0.55mg electrons
- (D) All of these



[Android App](#)



[iOS App](#)



[PW Website](#)

Answer Key

Q1 A
Q2 C
Q3 B
Q4 A
Q5 A

Q6 C
Q7 D
Q8 C
Q9 B
Q10 D



[Android App](#)



[iOS App](#)



[PW Website](#)

Hints & Solutions

Note: scan the QR code to watch video solution

Q1 Text Solution:

All homogenous mixture are called as solutions.

Video Solution:



Q2 Text Solution:

According to the law of conservation of mass, in ordinary chemical reactions, the total mass of reactants equals the total mass of products, so the overall mass remains **unchanged**.

Video Solution:



Q3 Text Solution:

Nuclear reactions can convert mass into energy ($E=mc^2$), so total mass isn't strictly conserved. In ordinary chemical (including endothermic or exothermic) reactions, total mass remains unchanged.

Video Solution:



Q4 Text Solution:

Option(A) directly illustrates that the total mass of the reactants ($12\text{gC} + 32\text{gO}_2 = 44\text{g}$) equals the total mass of product (44gCO_2), demonstrating the law of conservation of mass.

Video Solution:



Q5 Text Solution:

Thermal decomposition of CaCO_3 yields CaO and CO_2 only. Hence, if 10g CaCO_3 forms 5.6g CaO , the difference ($10 - 5.6 = 4.4\text{g}$) is the mass of CO_2 .

Video Solution:



Q6 Text Solution:

The law of multiple proportions requires two compounds made of *the same two elements* in different integer ratios. **Cu_2O** and **CuO** both contain copper and oxygen, but with differing Cu:O ratios ($2:1$ vs. $1:1$), illustrating that law.

Video Solution:



[Android App](#)

| [iOS App](#)

| [PW Website](#)

Q7 Text Solution:

- The Law of Multiple Proportions is defined as: "if two elements combine to form more than one compound, the mass ratios of the second element that combine with a fixed mass of the first element will always be ratios of minuscule whole numbers."
- Ratio of masses of oxygen in two oxides = $\frac{53.33}{36.36} \approx \frac{3}{2}$
- The oxygen atoms in two oxides bears a simple ratio by mass 3:2
- Hence it illustrates the Law of multiple proportions.

Video Solution:**Q8 Text Solution:**

Average atomic mass of two isotopes A_1 and A_2 is:

$$\begin{aligned} \text{Avg} &= \frac{\%A_1 + \%A_2}{100} \\ 10.81 &= \frac{80 \times 11.01 + 20A_2}{100} \\ 1081 &= 880.8 + 20A_2 \\ 20A_2 &= 1081 - 880.8 \\ 20A_2 &= 200.2 \\ A_2 &= 10.01 \end{aligned}$$

Video Solution:**Q9 Text Solution:**

Let % abundance of C-12 be x and $(100-x)$ be % abundance of C-13.

$$\begin{aligned} \frac{12x + 13(100-x)}{100} &= 12.011 \\ 12x + 1300 - 13x &= 1201.1 \\ x &= 98.9 \end{aligned}$$

Abundance of C-12 is 98.9%

Video Solution:**Q10 Text Solution:**

Number of electrons in 1 mole $N_A = 6.022 \times 10^{23}$ atoms, N_A is avagadro number.

Mass of 1 mole of electron = mass of 1 electron $\times N_A$

$$\begin{aligned} &= 9.1 \times 10^{-28} \times 6.022 \times 10^{23} = 5.48 \\ &\times 10^{-4} \text{ g} \simeq 0.55 \text{ mg} \end{aligned}$$

One mole of electron means All of these.

Video Solution:

[Android App](#)

| [iOS App](#)

| [PW Website](#)